



Hah Min Lew

PH.D. STUDENT · KOREA UNIVERSITY

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“Bridging AI research and scalable solutions for measurable real-world impact.”

Biography

Ph.D. student at the Visual & General Intelligence (VGI) Lab, Korea University, supervised by Prof. Gyeong-Moon Park. Current research focus lies in Vision-Language-Action (VLA) models, Vision-Language-Navigation (VLN) models, and multimodal generative models.

Prior to academia, professional experience includes over three years as an AI Researcher at Klleon AI Research. Key contributions involved photo-realistic 3D human generation, enabling lifelike avatar movements and lip-sync. Holds both M.S. and B.E. degrees from DGIST, with a strong foundation in Deep Learning and Signal Processing, advised by Prof. Jae Youn Hwang.

Skills

Programming	Python, Bash, MATLAB, C, Java
Frameworks	PyTorch, TensorFlow, Keras, Scikit-learn, Pytorch3D
DevOps	Docker, Containerd, Git
Back-end Basics	Node.js
Front-end Basics	HTML, CSS, JavaScript
Languages	Korean, English

Education

Korea University

PH.D. IN ARTIFICIAL INTELLIGENCE

- Advisor: Prof. Gyeong-Moon Park

Seoul, South Korea

Mar. 2026 - present

DGIST (Daegu Gyeongbuk Institute of Science and Technology)

M.S. IN ELECTRICAL ENGINEERING AND COMPUTER SCIENCE

- Advisor: Prof. Jae Youn Hwang

Daegu, South Korea

Mar. 2019 - Aug. 2022

DGIST (Daegu Gyeongbuk Institute of Science and Technology)

B.E. IN SCHOOL OF UNDERGRADUATE STUDIES

- Best Project Award

Daegu, South Korea

Mar. 2014 - Feb. 2019

Experience

Klleon AI Research

AI RESEARCHER

- Accelerated inference using a training-free diffusion sampler ($\uparrow 4.11\times$).
- Developed audio-driven 3DMM generation for virtual avatars with natural human-like expressions and movements. (LVE $\downarrow 27.5\%$, FDD $\downarrow 28.9\%$, MEE $\downarrow 27.1\%$, CE $\downarrow 24.1\%$, Diversity $\uparrow 17.7\%$)
- Developed a photorealistic head rendering model using Gaussian Splatting, outperforming 5 state-of-the-art models. (MSE $\downarrow 59.96\%$, PSNR $\uparrow 4.41dB$, SSIM $\uparrow 3.85\%$, LPIPS $\downarrow 38.16\%$)
- Developed a Head Swap AI model with 212.7x speedup, reducing annual GPU costs by 99.53% (from \$2.2M to \$10.5K).
- Built large-scale multimodal data pipelines (9.41M+ frames from in-the-wild videos).
- Integrated external APIs (NVIDIA Omniverse Audio2Face, OpenAI ChatGPT, TTS) into a streaming avatar system, achieving 25-28 FPS performance.

Seoul, South Korea

Aug. 2022 - Feb. 2026

Multimodal Biomedical Imaging and System Lab, DGIST

GRADUATE RESEARCHER

- Achievements: 6 SCIE publications, 7 international conferences, 9 projects, 4 patents, and 2 awards.
- Designed machine learning-based anomaly detection systems and generative models for biomedical imaging, focusing on multimodal, class-imbalance, and multi-task learning.
- Developed hardware-software integrated systems for application-specific use cases.
- Collaborative research with medical doctors from hospitals, including SNUH, SNUDH, and Yonsei Severance.

Daegu, South Korea

Mar. 2019 - Aug. 2022

Publications

[C5] AudioAvatar: Personalized Audio-driven Whole-body Talking Avatars

S. LEE, S. MOON, **HAH MIN LEW**, J.-S. KANG, AND G.-M. PARK. **CVPR 2026**

Co-Author

Jun. 2026

[C4] SMAGA: Secondary Motion-Aware 3D Clothed Gaussian Avatars from Monocular Videos

S. LEE, S. MOON, **HAH MIN LEW**, J.-S. KANG, AND G.-M. PARK. **ICLR 2026**

Co-Author

Apr. 2026

[C3] GeoAvatar: Adaptive Geometrical Gaussian Splatting for 3D Head Avatar

S. MOON*, **HAH MIN LEW***, S. LEE, J.-S. KANG, AND G.-M. PARK. **ICCV 2025**

First Author

Oct. 2025

[C2] Towards High-fidelity Head Blending with Chroma Keying for Industrial Applications

HAH MIN LEW*, S.-M. YOO*, H. KANG*, AND G.-M. PARK. **WACV 2025**

First Author

Mar. 2025

[C1] CSS-Net: Classification and Substitution for Segmentation of Rotator Cuff Tear

K. LEE, **HAH MIN LEW**, M. H. LEE, M. KANG, J. KIM, AND J. Y. HWANG. **ACCV 2022**

Co-Author

Dec. 2022

[J6] Multi-task and Few-shot Learning-based Fully Automatic Deep Learning Platform for Mobile Diagnosis of Skin Diseases

K. LEE, T. C. CAVALCANTI, S. KIM, **HAH MIN LEW**, D. H. SUH, D. H. LEE, AND J. Y. HWANG. **IEEE JBHI** (IF: 7.021)

Co-Author

Jan. 2023

[J5] Deep Learning-based Framework for Fast and Accurate Acoustic Hologram Generation

M. H. LEE, **HAH MIN LEW**, S. YOUN, T. KIM, AND J. Y. HWANG. **IEEE TUFFC** (IF: 3.267)

Co-Author

Dec. 2022

[J4] Speckle Reduction via Deep Content-Aware Image Prior for Precise Breast Tumor Segmentation in an Ultrasound Image

H. LEE, M. H. LEE, S. YOUN, K. LEE, **HAH MIN LEW**, AND J. Y. HWANG. **IEEE TUFFC** (IF: 3.267)

Co-Author

Sep. 2022

[J3] Intelligent Smartphone-based Multimode Imaging Otoloscope for the Mobile Diagnosis of Otitis Media

T. C. CAVALCANTI, **HAH MIN LEW**, K. LEE, S. LEE, M. K. PARK, AND J. Y. HWANG. **BIOMEDICAL OPTICS EXPRESS** (IF: 3.562)

Co-Author

Dec. 2021

[J2] Ultrasonic Blood Flowmeter with a Novel Xero Algorithm for a Mechanical Circulatory Support System

HAH MIN LEW, H. SHIN, M. H. LEE, S. YOUN, H. C. KIM, AND J. Y. HWANG. **ULTRASONICS** (IF: 4.062)

First Author

Aug. 2021

[J1] Forward-Looking Multimodal Endoscopic System Based on Optical Multispectral and High-Frequency Ultrasound Imaging Techniques for Tumor Detection

J. KIM, **HAH MIN LEW**, J. KIM, S. YOUN, H. A. FARUQUE, A. N. SEO, S. Y. PARK, J. H. CHANG, E. KIM, AND J. Y. HWANG. **IEEE TMI** (IF: 11.037)

Co-Author

Feb. 2021

Projects

Audio-driven 3D Facial Animation for Realistic Facial Expressions and Motion

Seoul, South Korea

PROJECT LEAD

Dec. 2024 - Feb. 2026

- Developed a 3D facial animation framework for lifelike facial expressions and motion driven by audio inputs.
- Constructed a large-scale paired dataset of audio and 3DMM parameters (6.81M+ frames).
- Achieved superior performances compared to the SOTA method (LVE $\downarrow$ 27.5%, FDD $\downarrow$ 28.9%, MEE $\downarrow$ 27.1%, CE $\downarrow$ 24.1%, Diversity $\uparrow$ 17.7%).
- Accelerated diffusion sampling process 4.11x speed-up while preserving qualitative performances.
- Used skills: Python, PyTorch, Git.

Real-time Expressive 3D Chat Avatar System

Seoul, South Korea

PROJECT LEAD

Apr. 2024 - Dec. 2024

- Integrated NVIDIA Omniverse Audio2Face, OpenAI ChatGPT, and TTS APIs into a streaming avatar dialogue system with 25-28 FPS performance.
- Designed an emotion message queue protocol to enable natural emotional transitions and realistic facial expressions in avatars.
- Optimized Numpy-to-Tensor conversion and computations for live streaming, achieving a 13.5% speed improvement.
- Used skills: Python, PyTorch, Docker, Containerd, Git.

High-performance Real-time Head Swapping System

Seoul, South Korea

PROJECT LEAD

Aug. 2022 - Apr. 2024

- Led the development of a state-of-the-art head swapping framework, including data preprocessing pipelines, multi-GPU training, and efficient inference mechanisms.
- Built a high-quality dataset from 15,354 videos of 3,592 identities, processing 2.6M frames.
- Achieved a 212.7x inference speedup (from 10s/frame to 47ms/frame), reducing GPU resource requirements by 99.53%.
 - Reduced annual GPU costs from \$2.2M+ (assuming 213 AWS EC2 g4dn.4xlarge instances) to approximately \$10.5K (using a single instance).
- Achieved significant performance improvements over the SOTA method:
 - Metrics: PSNR $\uparrow$ 55.5%, LPIPS $\downarrow$ 91.8%, L1 $\downarrow$ 88.8%, SSIM $\uparrow$ 21.8%
 - Inference speed: 60.57 FPS ($\uparrow$ 53.6%)
 - Computational efficiency: Parameters 8.92M ($\downarrow$ 63.4%), MACs $\downarrow$ 33.0%
- Used skills: Python, PyTorch, Docker, Git, JavaScript, HTML, CSS.

ADDITIONAL PROJECTS

Image-to-Image Translation for High-resolution Gastrointestinal Imaging PROJECT LEAD	Feb. 2021 - Sep. 2023
Multitask Learning-based Network for Rotator Cuff Tear Segmentation PROJECT MEMBER	Dec. 2021 - Dec. 2022
Low-voltage CMUT-based Ultrasound Imaging for Medibots PROJECT MEMBER	Sep. 2020 - Dec. 2022
AI-powered Smartphone Imaging for Early Dental Caries Detection PROJECT LEAD	Apr. 2020 - Feb. 2022
ML-based Smartphone Imaging for Otitis Media Diagnosis PROJECT MEMBER	Feb. 2020 - Jan. 2022
2021 Laboratory-specialized Start-up Leader University Project PROJECT MEMBER	Aug. 2021 - Jan. 2022
Smart Monitoring System for Hip Implants PROJECT MEMBER	Feb. 2019 - May. 2021
Optimized Biomedical Monitoring System with a Time-efficient Algorithm PROJECT LEAD	Mar. 2019 - Mar. 2021
Technical Commercialization Activity Support for Bio Society Leadership PROJECT MEMBER	May. 2020 - Dec. 2020
Multimodal Data Registration and Analysis for Tumor Detection PROJECT MEMBER	Mar. 2019 - Oct. 2020
Ultrasonic Capsule Endoscopy PROJECT MEMBER	Jun. 2019 - Jun. 2020

Patents

COMPUTING DEVICE FOR HEAD SWAPPING	<i>Application</i>
HAH MIN LEW , H. KANG, S.-M.YOO, G.-M.PARK (WO2025042068A1, KR1020230154188A)	Feb. 2025
MOBILE OTOSCOPE SYSTEM	<i>Patent</i>
J. Y. HWANG, T. C. CAVALCANTI, HAH MIN LEW (KR102648059B1)	Mar. 2024
ULTRASONIC BLOOD FLOW MEASURING APPARATUS AND METHOD THEREOF	<i>Patent</i>
J. Y. HWANG, HAH MIN LEW , H. C. KIM (KR102514633B1)	Mar. 2023
BLADDER MONITORING APPARATUS AND METHOD FOR CONTROLLING BLADDER MONITORING APPARATUS	<i>Patent</i>
J. Y. HWANG, M. H. LEE, HAH MIN LEW (KR102460829B1)	Oct. 2022
THREE-DIMENSIONAL DIAGNOSTIC SYSTEM	<i>Patent</i>
J. Y. HWANG, J. KIM, HAH MIN LEW (KR102379481B1)	Mar. 2022

Awards & Scholarships

AWARDS

2021	Outstanding Poster Award , 2021 Student Conference at DGIST	<i>Daegu, South Korea</i>
2021	Best Paper Award , 2021 Spring Conference at KOSOMBE	<i>Remote, South Korea</i>
2017	Best Project Award , 2016 Undergraduate Group Research Project (UGRP) Program at DGIST	<i>Daegu, South Korea</i>

SCHOLARSHIPS

2014-2022	Full Government Scholarships , Full tuition exemptions and school expenses support	<i>Daegu, South Korea</i>
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References

Gyeong-Moon Park

ASSOCIATE PROFESSOR

- Ph.D. in School of Electrical Engineering, KAIST, Seoul, South Korea. 2019.
- E-mail: gm-park@korea.ac.kr
- Office: Room #203B, Woo Jung Informatics Building

Seoul, South Korea

Mar. 2025 - present

Jae Youn Hwang

PROFESSOR

- Ph.D. in Biomedical Engineering, University of Southern California, Los Angeles, USA. 2009.
- E-mail: jyhwang@dgist.ac.kr
- Office: Room #413, E3 building

Daegu, South Korea

Sep. 2022 - present